

# Results

01 · PATH ANALYSIS

directed relationships among observed variables (path / mediation models)

Table 1. Structural paths

| Path                      | B     | SE   | z     | p     | Std. β | 95% CI      | R²  |
|---------------------------|-------|------|-------|-------|--------|-------------|-----|
| esg_score → norm_score    | −0.10 | 0.05 | −1.93 | .054  | −.10   | [−.20, .00] | .01 |
| norm_score → intent_score | 0.25  | 0.05 | 5.30  | <.001 | .26    | [.17, .35]  | .17 |
| esg_score → intent_score  | 0.35  | 0.04 | 7.82  | <.001 | .34    | [.26, .42]  | .17 |

Table 2. Indirect effects

| Path                                  | Estimate | SE   | boot 95% CI    | p    |
|---------------------------------------|----------|------|----------------|------|
| esg_score → norm_score → intent_score | −0.03    | 0.01 | [-0.06, -0.00] | .077 |

**Saturation:** The model is saturated (df = 0): it has zero degrees of freedom, so global fit indices are not informative and are suppressed. Estimated paths and effects below are still interpretable.

Figure. Path diagram

−

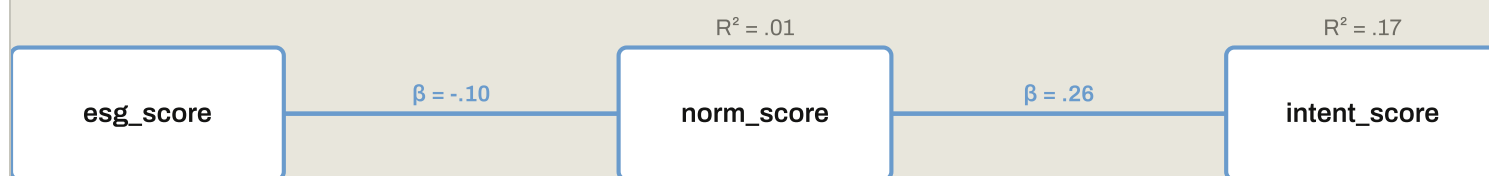
+

Fit

Draw path

Move

Delete



**95% CI.** The confidence interval for a structural path's estimate, or the bootstrap 95% CI for an indirect effect. Here, each structural path has its own 95% CI, and each indirect effect its own bootstrap 95% CI (see the tables above) - an interval excluding 0 indicates a credible effect.

**R<sup>2</sup>.** The variance explained in each endogenous (outcome) variable. Here, R<sup>2</sup> ranges from .01 to .17 across the 3 paths.

**Estimate.** The size of an indirect (mediated) effect - the product of the paths through a mediator. Here, indirect-effect estimates range from -0.03 to -0.03 across the 1 mediation chain(s).

### How to read this test

Path analysis estimates a system of regressions among observed variables at once, letting one variable be both an outcome and a predictor (mediation). Structural paths: each B is the unstandardized effect, Std.  $\beta$  the standardized effect, with z, p, and a 95% CI; R<sup>2</sup> is the variance explained in each endogenous variable. Indirect effects: the product of the paths through a mediator (e.g.  $X \rightarrow M \rightarrow Y$ ), judged by its bootstrap 95% CI - an interval excluding 0 indicates mediation. When df = 0 the model is saturated (it reproduces the data exactly), so fit indices are not informative and are omitted; only with extra constraints (df > 0) do global fit indices apply, and even then read RMSEA cautiously at small df / small N. All thresholds are heuristics, not pass/fail gates.

**APA template:** A path model fit to the observed variables; the indirect effect of X on Y through M was significant, bootstrap 95% CI excluding 0.

**Statistical basis:** Path model estimation (lavaan::sem, observed variables only) (Rosseel, Y. (2012). "lavaan: An R Package for Structural Equation Modeling." Journal of Statistical Software, 48(2), 1-36.); RMSEA interpreted cautiously at small df / small N for over-identified models (Kenny, D. A., Kaniskan, B., McCoach, D. B. (2015). "The performance of RMSEA in models with small degrees of freedom." Sociological Methods & Research, 44(3), 486-507.); Bootstrap percentile CIs for indirect (mediated) effects (MacKinnon, D. P., Lockwood, C. M., Williams, J. (2004). "Confidence limits for the indirect effect: Distribution of the product and resampling methods." Multivariate Behavioral Research, 39(1), 99-128.). Full references in the exported CITATIONS.txt.

